

SAM MAHDAVIAN, PhD, PMP, PMI-PBA

AI Solutions & Delivery Lead

Agentic systems that solve the business problem. Tested, governed, and safe in regulated production.

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PROFESSIONAL SUMMARY

AI solutions leader who turns ambiguous business problems into agentic systems that hold up in regulated production. I work as a single-threaded owner from discovery and pre-sales through reference architecture, build, and adoption, and I hold every system to one test: does it solve the business problem, is it evaluated against real cases, and is it safe to run? Across healthcare, financial services, transportation, retail, and manufacturing I have shipped **30+ systems and 110+ agents** that together produced **\$40M+ in revenue and \$500M+ in identified client savings**. I also co-founded and run an AI infrastructure-planning startup as CEO. 15+ years pairing LLM/RAG and AWS deployment engineering with executive-facing delivery and team leadership. PhD (ML / applied AI); PMP; PMI-PBA.

CORE COMPETENCIES

Solutions & Pre-Sales: enterprise AI delivery, solution architecture, discovery to design to deploy to adopt, requirements elicitation across sectors, pre-sales and POCs, buyer-persona and commercial market analysis, ROI and business-case framing, single-threaded ownership, executive enablement.

Agentic AI & GenAI: LangChain/LangGraph, Strands SDK, CrewAI, custom Python; multi-agent orchestration; RAG and vector DBs; guardrails, evals, and human-in-the-loop; prompt engineering; latency/cost/accuracy optimization; AWS Bedrock and AgentCore; Claude and OpenAI.

Production & MLOps: AWS (Bedrock, SageMaker, Lambda, Step Functions, S3, ECS/EKS), Docker, Kubernetes, CI/CD, IaC (Terraform/CDK), model serving and LLMops, observability, model-risk governance, IAM and cloud security, HITRUST/HIPAA, FinOps.

Leadership & Delivery: team building and mentorship (orgs to ~25), roadmap and solution ownership, GTM, pricing and packaging, Agile/Scrum and SAE, storytelling with data, hiring and talent development.

PROFESSIONAL EXPERIENCE

AArete LLC

Orlando, FL · 2022 - Present

Director, Data Science & AI (2024 - Present) · Manager (2023 - 2024) · Senior Data Scientist (2022)

- Designed the architecture and led the build of **Doczy.ai**, a multi-tenant GenAI platform on AWS (retrieval, prompt orchestration, agentic tool-calling, dynamic model selection, governance, observability). Scaled it to **2.5M+ contracts and 442B+ tokens at 99%+ accuracy** under HITRUST/HIPAA, generating **\$15M+ in revenue and \$330M+ in client savings**. Architecture featured on the AWS Architecture Blog; issued a U.S. patent in 2025.
- Built **30+ systems and 110+ agents** that each solve a specific business problem, most now in production, on LangChain/LangGraph, Strands SDK, and custom Python over AWS Bedrock and AgentCore. Across the portfolio these systems produced **\$40M+ in revenue and \$500M+ in identified client savings**. Every one is evaluated against real cases before release, tuned for latency, cost, and accuracy, and built human-in-the-loop and audit-first.
- Run discovery and pre-sales as a single-threaded owner across 10+ Fortune 500 engagements. I sit with executives and business teams, map the real pain by sector, and convert ambiguous problems into structured SOWs and roadmaps. Recently won a **\$700K+ engagement with an equipment-finance lender**, plus **\$2.5M+** in competitive government and industry RFPs.
- Own production deployment end to end: containerized services (Docker, ECS/EKS), CI/CD, IaC, Bedrock and SageMaker model serving, guardrails and evals, observability, IAM and security, and cost controls, so the systems run reliably and safely at client scale.
- Authored the firm's AI-governance framework (model-risk management, bias and drift monitoring, explainability, prompt-injection defense, confidence-scored human-in-the-loop, incident response) and a four-stage maturity model that gates every system to production before it ships.
- Lead and mentor ~15 AI scientists and engineers across backend, frontend, and DevOps, plus a business-translation layer that turns stakeholder needs into specs and prompts. Run the firm-wide AI delivery program: **~\$1.3M budget, 10+ concurrent workstreams**, a six-gate value framework, and value-based pricing that protects margin as delivery cost falls.
- Lead executive AI-enablement: a one-day workshop for 40+ leaders at a Fortune 500 manufacturer produced a **30% productivity increase and 20+ prioritized use cases**. Earlier (Manager / Sr. Data Scientist), built an AI/ML Accelerator that cut modeling cycles ~75% and an ML suite proving a **4.1% registration uplift worth \$9.6M** in new annual revenue.

Brane Mobility (SPLUSM LLC)

Orlando, FL · 2020 - Present

Co-Founder & CEO

- Co-founded and lead **Brane Mobility** as Co-Founder and CEO, a cloud-native B2B SaaS that turns raw mobility and urban data into multi-scenario, stakeholder-ready insights in minutes for city planners, public agencies, and enterprises. Built and led a **~25-person** cross-functional org (GenAI and data scientists, data engineers, full-stack developers, UX/UI) and **raised \$625K**.
- Architected the platform as two integrated systems: an advanced **data-management layer** (a four-level dataset-to-KPI hierarchy with automated ETL, data validation, a custom KPI formula builder, and user-data integration) and an **end-to-end five-step analysis workflow** (project definition across 18 problem types, hotspot identification, multi-objective optimization, sensitivity analysis, and automated Word, PDF, and slide reporting).
- Designed the **AI Decision Engine**: neural-network forecasting, genetic-algorithm multi-objective optimization, sensitivity analysis, and an AWS Bedrock Gen AI copilot that guides users in natural language. Ran multi-city pilots showing **~18% operational efficiency gains**, and am now extending the engine toward grid and large-load siting with a USDOT FTA SBIR and a seed raise in motion.

UCF College of Engineering & Computer Science

Orlando, FL · 2017 - 2020

Lead Data Scientist & Researcher

- Delivered two production applications with end-to-end data pipelines used by the **U.S. Department of Transportation** for traffic and cost forecasting; secured competitive USDOT and NSF funding. Led applied-AI research in forecasting, optimization, and simulation, producing **20+ peer-reviewed publications**.

Stratus International Contracting Co.

2011 - 2017

Senior Data Analyst

- Built a PMO optimization framework (hybrid genetic algorithm plus constraint-based simulation) and stood up the firm's first PMI-aligned PMO, cutting **21% of cost and 32% of time** across major projects in three years.

SELECTED PRODUCTION AI SOLUTIONS (BUSINESS OUTCOME FIRST)

- **Recovered millions in wrongly-paid claims** for a top-5 health payer. A 9-agent pipeline ranks recovery and generates 200+ Snowflake-ready SQL queries per policy. Analysis went from 10 to 20 days down to **90 minutes**; reused across payers.
- **Protected margin on thousands of daily financing deals** for an equipment-finance lender. A dynamic pricing engine (XGBoost risk scoring, reinforcement-learning optimization, SHAP explainability) delivered a **15 to 20% margin improvement and 60% faster decisions**, explainable for compliance. (\$700K+ engagement.)
- **Surfaced spend savings across 14,765 suppliers and 47,489 parts** for a Fortune 500 airline. A 10-agent supplier-intelligence pipeline plus a React command center cut prep from weeks to 2 days, **98% less effort, +20% accuracy**.
- **Cut network-analysis cost from \$5,760 to \$803 per engagement** for a state Medicaid program with a 7-agent LangGraph cognitive loop, taking a five-day analysis down to **8 hours**.
- **Made underwriting transparent and regulator-ready** with an 11-agent system that scores 11 risk dimensions under a Chief Underwriter agent (XGBoost + SHAP, every decision explained). Open source.
- **Built contract review you can audit**: a deterministically-orchestrated engine rates every clause P/A/D/U with an independent QC critic, citation-integrity enforcement, prompt-injection isolation, and human sign-off before release.
- **Caught the fraud rings rules miss** with a graph-neural-net plus LangGraph multi-agent investigation over a 3D transaction graph. Open source.
- **Answered procurement and policy questions in under 3 seconds** with a hybrid-RAG knowledge hub (>92% accuracy, 100% cited, cite-or-refuse), and produced Medicaid Fee-Schedule BRDs in ~3.5 minutes versus days, scaling to 50 states.
- **Turned raw SEC filings into analyst-ready answers** with an MCP tool-calling SaaS: CrewAI analyst agents call Model Context Protocol tools (filings, market metrics) under a governance layer, with citations on every claim.
- **Cut LLM cost ~40% and made agents debuggable** by instrumenting a multi-agent crew with full LangFuse tracing, so every step, token, and dollar is observable in production.

EDUCATION & CERTIFICATIONS

Ph.D., Engineering, University of Central Florida. Dissertation: machine-learning algorithms for forecasting (ML / applied-AI methods).
M.S., Engineering, UCF. Executive Education, Business Administration, Harvard Business School.

Certifications: PMP · PMI-PBA · Certified SAFe Practitioner · Advanced Data Science Fellowship (The Data Incubator) · **Claude Certified Architect, Anthropic** (in progress, 2026) · **AWS Solutions Architect Associate** (in progress, 2026).

PUBLICATIONS, PATENTS & RECOGNITION

Patent & features: U.S. Patent (2025), GenAI contract-intelligence method (Doczy.ai) · featured on the **AWS Architecture Blog** · founder of Modern AI Academy, teaching how agentic AI is built and shipped.

Research: **20+ peer-reviewed publications, 443 citations, h-index 9** (Google Scholar). Most-cited: drivers and barriers to connected, automated, shared, and electric vehicles (IEEE Access, 130+ citations); automated ML pipeline for traffic prediction (MDPI AI); data-driven highway-cost modeling (ASCE). \$2.5M+ in competitive research funding (USDOT, NSF).